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Deep Learning and Its Applications (CO3133)

Semester 261 – Class CC01 – Group G-02

Assignment 2 – Milestone 2

M2 Draft

Tomato Leaf Disease Classification

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1 EDA and data description

Source: `Assignment2/01_eda.ipynb`. The notebook checks the dataset against the M1 proposal and tests the leakage-prevention requirement.

The Tomato Leaves Dataset [1] combines tomato images from PlantVillage [2] and PlantDoc [3]. Each image has one label: seven diseases or a healthy leaf.

1.1 Read the dataset

Use the split published in the dataset repository (revision `c029791`). Images are listed by folder; the label is the folder name.

Class	train	validation	test	Total
Bacterial spot	1930	235	241	2406
Early blight	966	131	117	1214
Late blight	1710	210	209	2129
Leaf mold	999	120	125	1244
Septoria leaf spot	1771	210	223	2204
Tomato yellow leaf curl virus	4930	624	625	6179
Tomato mosaic virus	509	66	59	634
Healthy	1589	200	197	1986
Total	14404	1796	1796	17996

Table 1: Images per class and partition.

The published split holds 17,996 images: 14,404 for training and 1,796 each for validation and test. These counts match the M1 proposal. Tomato yellow leaf curl virus is the largest class (6,179 images, 4,930 in training) and Tomato mosaic virus the smallest (634 images, 509 in training).

1.2 Class distribution and imbalance

Compare the share of each class in the three partitions. A partition that differs strongly from the training set would make validation or test scores hard to compare.

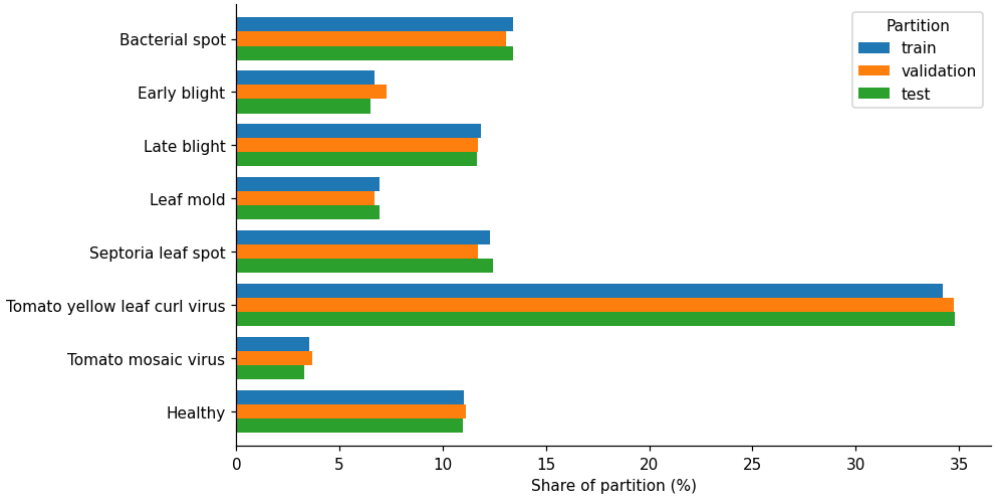


Figure 1: Share of each class in the three partitions.

	Value
Largest class	Tomato yellow leaf curl virus
Largest count	6179
Smallest class	Tomato mosaic virus
Smallest count	634
Imbalance ratio	9.7

Table 2: Class imbalance.

The largest class is 9.7 times the size of the smallest. The three partitions have almost the same class shares, so validation and test scores describe the same label mix as training. Accuracy alone would be dominated by the large virus class, so the experiments report macro-F1 as well and use it to select checkpoints.

1.3 Original image size and file format

Read only the image headers to learn the original resolution before the resize to 224×224.

	width	height
count	17996	17996
mean	285.9	279.8
std	223.6	186.9
min	115	85
25%	256	256
50%	256	256
75%	256	256
max	6000	6000

width	height	images
256	256	16889
224	224	317
800	600	23
1600	1200	19
1024	768	15
1200	1600	13

format	mode	images
JPEG	RGB	17995
PNG	RGBA	1

Table 3: Image size summary, most common sizes, and file formats.

17,995 files are RGB JPEG and one is an RGBA PNG, which the loader converts to RGB. 16,889 images (93.9%) are 256×256; 317 are 224×224, and the rest range from 115×85 to 6000×6000. Resizing every image to 224×224 therefore leaves most of the data almost unchanged, but it stretches the minority of non-square images.

1.4 Image sources

File names are anonymised (`image (N).JPG`), so the source cannot be read from the name. PlantVillage photographs are 256×256 crops, while PlantDoc images are web photographs of mixed size. Use the image size as a rough source label.

class	256 x 256 (PlantVillage-style)	Other size (PlantDoc / web-style)	Other share (%)
Bacterial spot	2274	132	5.5
Early blight	1108	106	8.7
Late blight	2004	125	5.9
Leaf mold	1110	134	10.8
Septoria leaf spot	1995	209	9.5
Tomato yellow leaf curl virus	5990	189	3.1
Tomato mosaic virus	539	95	15.0
Healthy	1869	117	5.9
train	13457	947	
validation	1718	78	
test	1714	82	

Table 4: Images by size group, per class and per partition.

File names are anonymised, so the size rule is only a rough proxy for the source. Under that rule, 6.6% of training images (947) have a size other than 256×256, against 4.3% of validation and 4.6% of test images. The share of such images varies by class, from 3.1% for Tomato yellow leaf curl virus to 15.0% for Tomato mosaic virus. A class may therefore partly be recognised by image style rather than by disease, and Tomato mosaic virus, the smallest class, has the largest share of web-style photographs.

1.5 Representative images

Show one random training image per class at the 224×224 model input size.

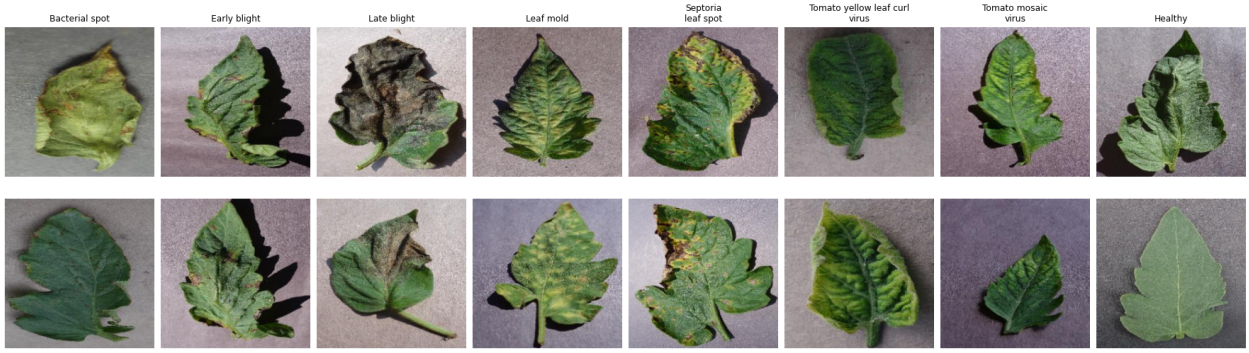


Figure 2: Random training images per class (seed 42).

Most sampled images show a single leaf on a plain grey background. The disease cues are small: spots, yellow areas, curled edges or dark patches. Several classes look alike at this size, for example Early blight, Septoria leaf spot and Bacterial spot, so fine texture will matter for the models.

1.6 Colour and brightness

Summarise the mean RGB value of every training image by class. Disease classes may differ in colour, which a small CNN can exploit.

class	R	G	B	Brightness
Bacterial spot	102.4	109.7	93.3	101.8
Early blight	109.3	114.0	100.4	107.9
Late blight	113.2	115.4	101.8	110.1
Leaf mold	111.4	117.8	111.7	113.7
Septoria leaf spot	129.5	132.4	117.0	126.3
Tomato yellow leaf curl virus	111.8	119.7	89.9	107.2
Tomato mosaic virus	116.9	118.1	114.0	116.3
Healthy	114.3	119.6	109.8	114.6
Training mean (0–1)	0.4438	0.4660	0.3936	

Table 5: Mean RGB value per class (0–255) and the training mean (0–1).

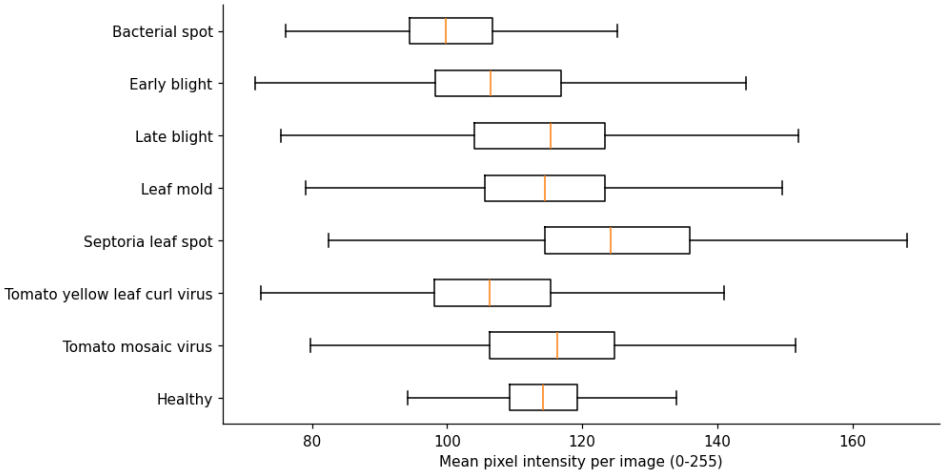


Figure 3: Mean pixel intensity per image (0–255), by class.

Mean colour differs by class. Septoria leaf spot images are the brightest (126.3) and Bacterial spot the darkest (101.8). Tomato yellow leaf curl virus has the lowest blue value (89.9), against 111.7 for Leaf mold and 114.0 for Tomato mosaic virus. Colour therefore carries some class information that a small CNN can use, but it may also reflect lighting or background differences between sources. The training mean (0.444, 0.466, 0.394) is close to the ImageNet mean used for normalisation.

1.7 Duplicates and leakage between partitions

The M1 proposal requires that an original photograph and its copies stay in one partition. Test this on the published split in two ways: byte-identical files, and visually near-identical images found with a 64-bit difference hash (Hamming distance of at most 4).

	Count
Files without an identical copy	17938
Duplicate groups	29
Distinct contents	17967
Groups spanning two or more partitions	3

Table 6: Byte-identical files (MD5).

17,938 files have no identical copy. The other 58 files form 29 duplicate groups of two byte-identical files, so the dataset holds 17,967 distinct contents. In 3 of those groups the two copies sit in different partitions. Exact copies are rare, so they are not the main leakage risk.

Partition	**Images**	**Identical hash (d=0)**	**Near duplicate (d≤4)**	**Share (%)**
validation	1796	5	88	4.9
test	1796	3	88	4.9
train (within itself)	14404	719	2873	19.9

Table 7: Images with a near-identical training image.

Near duplicates are much more common than exact copies. 88 validation images (4.9%) and 88 test images (4.9%) have a training image within Hamming distance 4. Inside the training set, 2,873 images (19.9%) have another training image

that close. The published split therefore does not fully keep each photograph and its copies in one partition, which the M1 proposal assumed.

Partition	Near duplicates	Same label as train match
validation	88	55
test	88	63

Table 8: Do the near duplicates agree on the label?

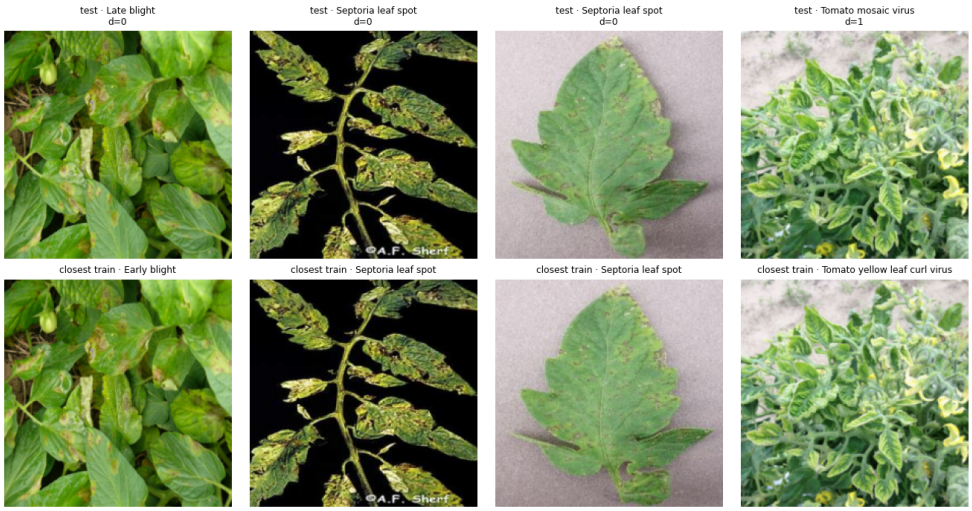


Figure 4: The four closest test images (top) and their closest training images (bottom).

Only 55 of the 88 validation matches and 63 of the 88 test matches carry the same label as the training image. The four closest test pairs are clearly the same photograph (distance 0 or 1), and in two of them the label differs: one image is Late blight in test and Early blight in training, another is Tomato mosaic virus in test and Tomato yellow leaf curl virus in training. This is evidence of duplicated photographs and some conflicting labels. Only the closest pairs have been checked by eye; with a threshold of 4 some of the 176 matches may be merely similar images, so the counts are an upper bound on true duplicates.

1.8 Inspect preprocessing

Show one training image before preprocessing, after the 224×224 resize, and after the two random flips used for training augmentation. Normalisation uses the ImageNet mean and standard deviation, which the pretrained ResNet-18 expects.

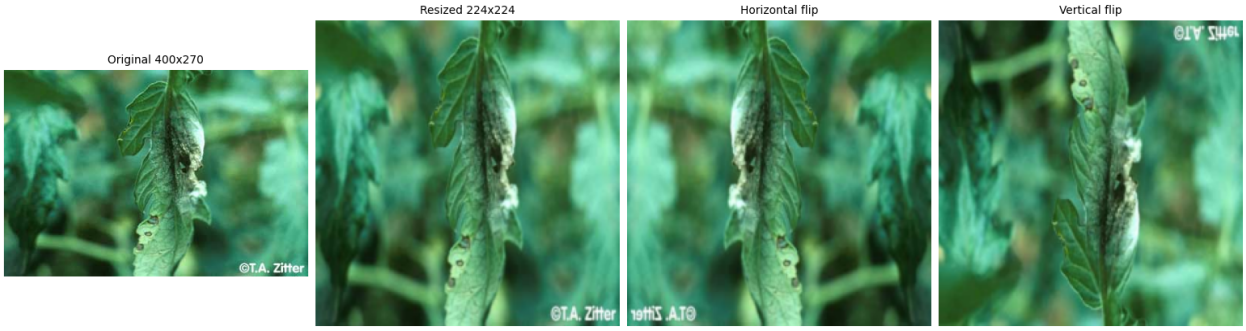


Figure 5: Original, resized, horizontally flipped and vertically flipped image.

Channel	Mean	Std
R	−0.900	0.620
G	0.330	0.850
B	−0.191	0.796

Table 9: Channel statistics of this image after normalisation.

The resized image keeps the leaf visible, and flips produce plausible views because a leaf has no fixed orientation. After normalisation the channels of this single image do not have zero mean and unit variance, because the statistics come from ImageNet, not from this image.

1.9 EDA conclusions

- **Split and size.** The data match the M1 proposal: 17,996 images in 8 classes and a 14,404 / 1,796 / 1,796 split.
- **Imbalance.** The class ratio is 9.7 to 1, so accuracy and macro-F1 can differ. Checkpoints are selected on validation macro-F1, and per-class results are examined in the results notebook.
- **Leakage.** Exact copies are rare, but about 5% of validation and test images have a near-duplicate in the training set, and some of those carry a different label. The M1 leakage-prevention rule is not met by the published split. For this draft the published split is kept so that results stay comparable with the proposal; the effect on the test score is measured in the results notebook.
- **Source style.** About 6% of images look like web photographs rather than lab crops, and the share differs by class. This is a possible shortcut for the models and a reason to look at errors by source.
- **Preprocessing.** Resize to 224×224, ImageNet normalisation, and random horizontal and vertical flips for training only. The same input pipeline serves the small CNN and ResNet-18.

2 Results

Source: `Assignment2/02_results.ipynb`. The notebook reports the preliminary metrics of the M2 Draft: a small CNN baseline trained from scratch and an ImageNet-pretrained ResNet-18 with a frozen backbone and with full fine-tuning. All models use one seed (42) and the published train / validation / test split.

2.1 Model definitions

- **CNN:** four convolution blocks (16, 32, 64, 64 channels) with batch normalisation, global average pooling, dropout 0.3 and a linear layer. No pretrained weights.
- **ResNet18Frozen:** ImageNet-pretrained [5] ResNet-18 [4]; only the new eight-class linear layer is trained, and the backbone stays in evaluation mode.
- **ResNet18FineTune:** the same network with all layers trained at a ten times lower learning rate.

2.2 Saved runs

Model	Seed	Parameters	Trainable	Epochs run	Selected epoch	Learning rate
CNN	42	61384	61384	12	12	0.0010
ResNet18Frozen	42	11180616	4104	6	5	0.0010
ResNet18FineTune	42	11180616	11180616	4	2	0.0001

Model	Median epoch seconds
CNN	56
ResNet18Frozen	194
ResNet18FineTune	526

Table 10: Saved runs.

The small CNN has 61,384 parameters. ResNet-18 has 11.2 million, of which the frozen variant trains only the 4,104 parameters of the new head. The CNN selected its last epoch (12 of 12), so it had not stopped improving; the budget, not convergence, ended its training. Frozen ResNet-18 selected epoch 5 of 6, and the fine-tuned model selected epoch 2 of 4. One epoch costs about 56 s for the CNN, 194 s for the frozen model and 526 s for full fine-tuning on the 16-thread CPU. The total fit time stored for the fine-tuned run is not usable, because its fourth epoch includes a pause of the computer (7,888 s); the median epoch time is used instead.

2.3 Validation comparison

Validation macro-F1 selected the checkpoint of every run.

Model	Loss	Accuracy	Macro-F1
CNN	0.2330	0.9265	0.8991
ResNet18Frozen	0.2911	0.9148	0.8806
ResNet18FineTune	0.0909	0.9733	0.9594

Table 11: Validation results.

Fine-tuned ResNet-18 leads on validation (accuracy 0.973, macro-F1 0.959). The CNN (0.927, 0.899) is ahead of the frozen backbone (0.915, 0.881).

2.4 Held-out test results

The test set was used once per model, after the checkpoint had been selected.

Model	Loss	Accuracy	Macro-F1	Inference ms/image
CNN	0.2383	0.9220	0.8955	6.3545
ResNet18Frozen	0.2938	0.9120	0.8754	11.4733
ResNet18FineTune	0.0876	0.9705	0.9547	10.8878

Table 12: Test results.

The test ranking matches validation. Fine-tuned ResNet-18 reaches 0.971 accuracy and 0.955 macro-F1, against 0.922 and 0.896 for the CNN and 0.912 and 0.875 for the frozen model. The gap between the fine-tuned model and the other two (about 5 points of accuracy) is large for one seed, but the gap between the CNN and the frozen model is about 1 point (roughly 18 images), which one seed cannot separate. The inference times are single measurements on a busy CPU and include data loading and normalisation, so they are only indicative; the frozen model, for instance, was timed at 17.0 ms per image on validation and 11.5 ms on test.

2.5 Learning curves

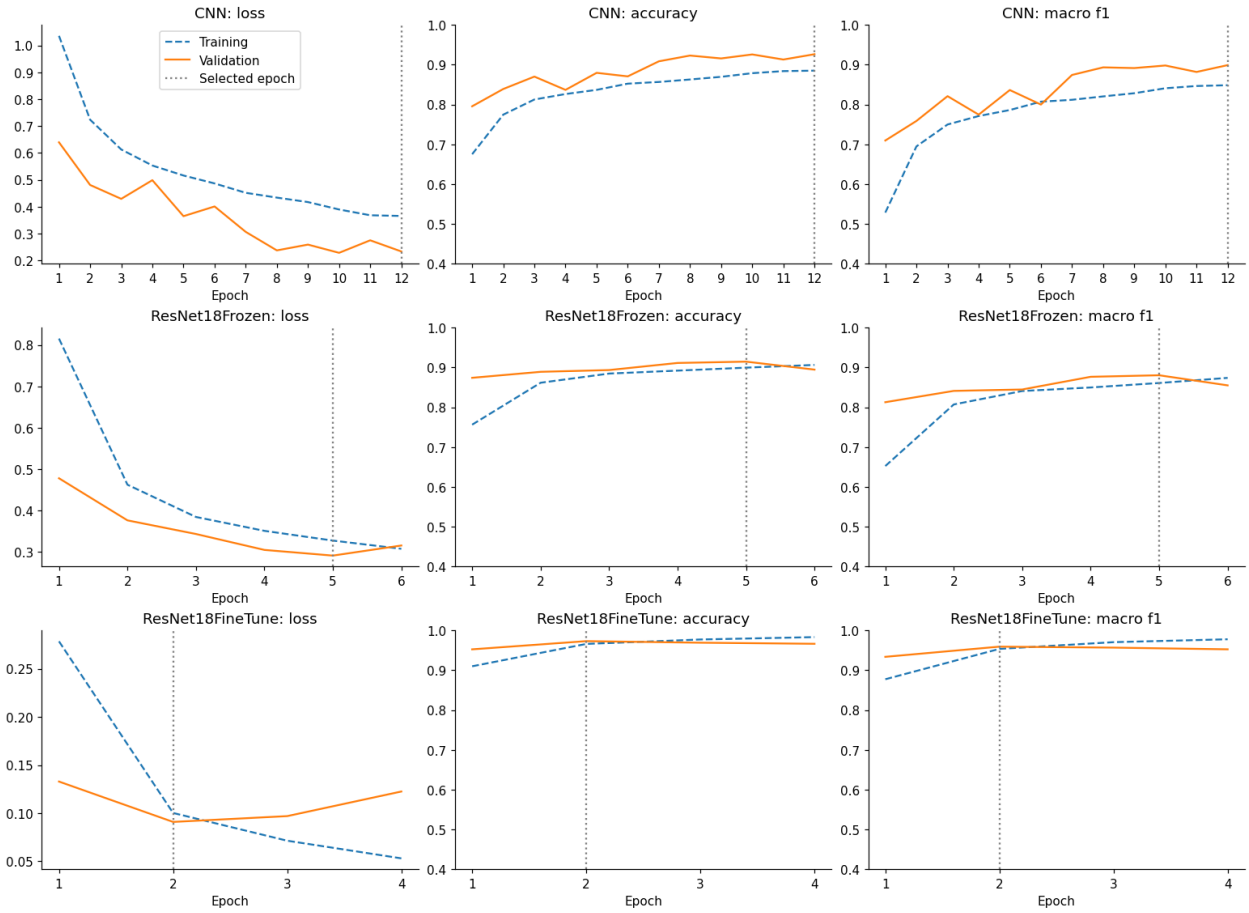


Figure 6: Training (dashed) and validation (solid) curves; the dotted line marks the selected epoch.

Validation scores sit above training scores for the CNN because training metrics are measured during the epoch with random flips and dropout. The CNN curves are still rising at epoch 12. For the frozen model, validation macro-F1 peaks at epoch 5 and drops at epoch 6, which shows how noisy this metric is when some classes have few validation images. The fine-tuned model reaches its best validation loss at epoch 2; after that the training loss keeps falling and the validation loss rises, so it starts to overfit within a few epochs at this learning rate.

2.6 Per-class results

Test F1 of each class. Classes are sorted in the dataset order.

Class	CNN	ResNet18Frozen	ResNet18FineTune	Test images
Bacterial spot	0.921	0.902	0.974	241
Early blight	0.775	0.765	0.938	117
Late blight	0.870	0.868	0.954	209
Leaf mold	0.895	0.825	0.921	125
Septoria leaf spot	0.910	0.881	0.963	223
Tomato yellow leaf curl virus	0.967	0.978	0.994	625
Tomato mosaic virus	0.857	0.821	0.909	59
Healthy	0.970	0.962	0.985	197

Table 13: Test F1 per class.

Fine-tuned ResNet-18 has the highest F1 for every class. The hardest classes for the baseline models are Early blight (0.775 for the CNN, 0.765 frozen) and Tomato mosaic virus (0.857, 0.821, only 59 test images); fine-tuning raises Early blight to 0.938 but Tomato mosaic virus only to 0.909. Tomato yellow leaf curl virus, the largest class, is easiest for both ResNet-18 models (0.978 and 0.994); for the CNN, Healthy is slightly higher (0.970 against 0.967).

2.7 Test confusion matrices

Rows are true classes, normalised to 1; columns are predicted classes.

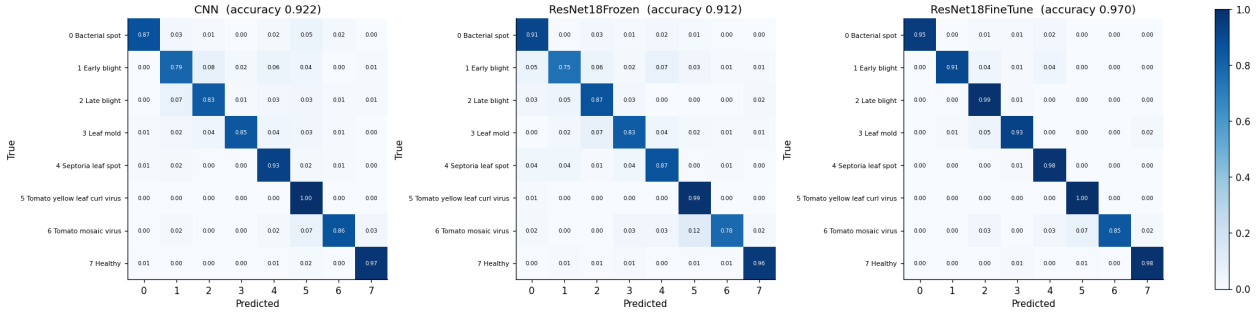


Figure 7: Test confusion matrices.

Most errors fall between visually similar diseases. Early blight is confused with Late blight and Septoria leaf spot by the CNN (8% and 6% of its images), and Tomato mosaic virus with Tomato yellow leaf curl virus by all three models (7% for the CNN and the fine-tuned model, 12% for the frozen model). Fine-tuning removes most of the Early blight and Late blight confusion but leaves the mosaic-versus-curl-virus confusion.

2.8 Effect of near-duplicate test images

The EDA found test images that have a near-identical image in the training set. Compare accuracy on those images with accuracy on the rest. The near-duplicate rule is the same as in the EDA: a 64-bit difference hash and a Hamming distance of at most 4 to any training image.

Model	Near-dup. images	Accuracy on them	Other images	Accuracy on the rest	Overall
CNN	88	0.9773	1708	0.9192	0.9220
ResNet18Frozen	88	0.9205	1708	0.9116	0.9120
ResNet18FineTune	88	0.9773	1708	0.9701	0.9705

Table 14: Test accuracy on near-duplicate and other images.

Near-duplicate images are easier than the rest for the CNN (0.977 on 88 images against 0.919 on the other 1,708), which is consistent with the model having seen a copy during training. For the two ResNet-18 models the difference is under 1 point. Removing the near-duplicates changes overall accuracy by less than 0.3 points for every model, so the leakage found in the EDA does not change the ranking of the models. With only 88 images, one image is worth 1.1 points, so these differences are indicative only.

2.9 Confident errors

For the model with the best validation macro-F1, show the test images it got wrong with the highest confidence.

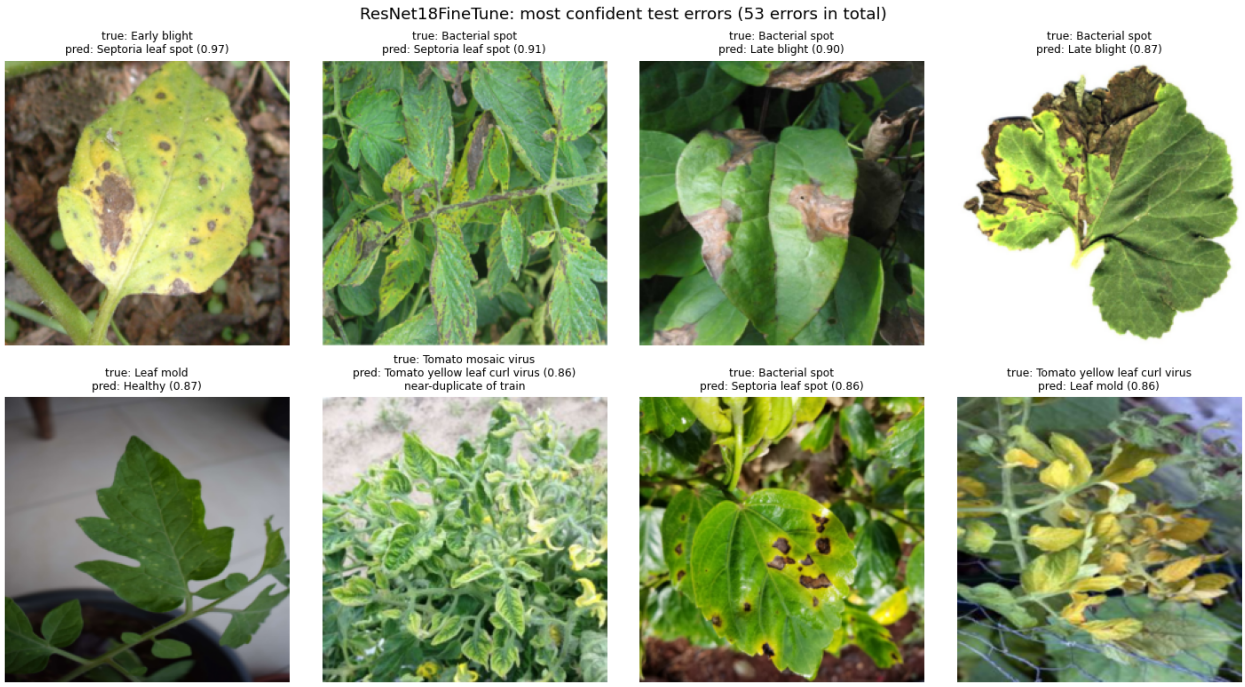


Figure 8: Most confident test errors of ResNet18FineTune.

	Count
Errors	53
Errors that are near-duplicates of a training image	2
Confident errors (≥ 0.9)	2

Table 15: Error counts of ResNet18FineTune.

The fine-tuned model makes 53 errors on the test set; only 2 of them are near-duplicates of training images and only 2 have a confidence of 0.9 or higher. The most confident errors are field photographs with several leaves or a cluttered background. At least two of the eight images shown (the fourth and the seventh) do not look like tomato leaves, which suggests label noise in the web-sourced part of the data; this has not been checked against the source.

2.10 Errors by image source

The EDA used the original image size as a rough source label: 256×256 images look like PlantVillage crops, and other sizes look like web photographs. Compare error rates for the two groups on the test set.

Model	256 x 256 images	Accuracy (256 x 256)	Other-size images	Accuracy (other sizes)
CNN	1714	0.9574	82	0.1829
ResNet18Frozen	1714	0.9411	82	0.3049
ResNet18FineTune	1714	0.9930	82	0.5000

Table 16: Test accuracy by image-size group.

The size proxy separates the data sharply. On the 1,714 test images of 256×256, accuracy is 0.957 (CNN), 0.941 (frozen) and 0.993 (fine-tuned). On the 82 images of other sizes it is only 0.183, 0.305 and 0.500. For the fine-tuned model that means about 41 of its 53 errors (77%) come from the 4.6% of test images that are web-style. The size rule is a rough proxy, and the 82 images are too few for precise estimates, but the gap is far larger than any difference between models.

2.11 Discussion

- **Pretrained fine-tuning is clearly best in this draft.** Full fine-tuning of ResNet-18 gives 0.955 test macro-F1, against 0.896 for the CNN trained from scratch. It also reaches its best validation score after only two epochs.
- **A frozen ImageNet backbone did not beat the small CNN.** The frozen model scored 0.875 macro-F1 and the CNN 0.896. With one seed and a 1-point accuracy difference this is not a firm ranking. A likely reason is that ImageNet features are not tuned to fine leaf texture, but this has not been tested.
- **The comparison is not at equal compute.** The CNN ran 12 epochs and was still improving, the frozen model 6 and the fine-tuned model 4. The baseline is probably under-trained, so the gap to the fine-tuned model may shrink with a longer run.
- **The hard part is the domain, not the architecture.** All models are above 94% accuracy on 256×256 lab-style images and between 18% and 50% on the small set of web-style photographs.
- **Leakage is present but does not change the ranking.** Near-duplicate test images are slightly easier for the CNN, but removing them moves overall accuracy by less than 0.3 points.
- **Limits of this draft.** One seed per model, short training budgets and no hyper-parameter search.

References

[1] L. Perinello and S. Giro, "Tomato Leaves Dataset," Hugging Face dataset repository, 2024. A merge of PlantVillage and PlantDoc, tomato-leaf classes only. https://huggingface.co/datasets/lorenzoxi/vcs-tomatoes-dataset. Copy used in this work: https://huggingface.co/datasets/shreemunale/tomato-leaves-dataset, revision c029791, CC BY 4.0.

[2] D. P. Hughes and M. Salathé, "An open access repository of images on plant health to enable the development of mobile disease diagnostics," *arXiv preprint arXiv:1511.08060*, 2015. https://arxiv.org/abs/1511.08060

[3] D. Singh, N. Jain, P. Jain, P. Kayal, S. Kumawat, and N. Batra, "PlantDoc: A dataset for visual plant disease detection," in *Proceedings of the 7th ACM IKDD CoDS and 25th COMAD*, pp. 249–253, 2020. https://arxiv.org/abs/1911.10317

[4] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016. https://doi.org/10.1109/CVPR.2016.90

[5] O. Russakovsky et al., "ImageNet large scale visual recognition challenge," *International Journal of Computer Vision*, vol. 115, no. 3, pp. 211–252, 2015. https://doi.org/10.1007/s11263-015-0816-y